



Vidyavardhini's College of Engineering and Technology
Department of Artificial Intelligence & Data Science

AY: 2024-25

Class:	SE	Semester:	III
Course Code:	CSL304	Course Name:	OBJECT ORIENTED PROGRAMMING JAVA

Name of Student:	SHRUTI GAUCHANDRA
Roll No. :	15
Assignment No.:	05
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	3
Legibility	2	2
Total	10	10

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : MS. NEHA RAUT
Signature : 
Date : 

Q1. A user wants to book the last available ticket of class Movie. Create two threads of the object of this class, created at two counters and book the ticket for one of the customers while display "House Full" for the other customer.

→ Code:

```
import java.io.*;
import java.util.*;
class Movie
{
    private int availableTicket = true;

    public synchronized void bookTicket(String
                                         customerName)
    {
        if (available Ticket)
        {
            System.out.println (CustomerName +
                                "Successfully booked the last ticket");
            availableTicket = false;
        }
        else
        {
            System.out.println ("House full for"
                                + CustomerName);
        }
    }
}
```

```

class TicketCounter extends Thread
{
    private Movie movie;
    private String customerName;
    public TicketCounter (Movie movie,
                          String customerName)
    {
        this.movie = movie;
        this.customerName = customerName;
    }
}

```

```

@Override
public void run()
{
    movie.bookTicket (customerName);
}
}

```

```

public class MovieBookingApp
{
    public static void main (String args[])
    {
        Movie movie = new Movie();
        TicketCounter counter1 = new TicketCounter(
            movie, "Customer 1");
        TicketCounter counter2 = new TicketCounter(
            movie, "Customer 2");

        counter1.start();
        counter2.start();
    }
}

```

```
}  
}
```

Output:

Customer 1 successfully booked the last ticket

House full for Customer 2

Q2. Implement a program to perform division of two numbers accepted from user. Handle the ID Exception, Number format exception and also handle the divide by zero exception using multiple try-catch block.

→ Code:

```
import java.io.*;
```

```
public class DivisionProgram  
{
```

```
    public static void main (String args[])  
    {
```

```
        BufferedReader reader = new BufferedReader (new  
            InputStreamReader (System.in));
```

```
        int num1 = 0, num2 = 0;
```

```
        try  
        {
```

```
            System.out.print ("Enter the first number:");
```

```
            try {
```

```
                num1 = Integer.parseInt (reader.readLine());
```

```
                // May throw Number format exception
```

```
            }  
            catch (NumberFormatException e)
```

```
            {
```

```
                System.out.println ("Invalid input for the  
                    first number. Please enter a valid  
                    integer.");
```

```
                return;
```

```

    }
    System.out.print("Enter the second number:");
    try {
        num2 = Integer.parseInt(reader.readLine());
    }
    catch (NumberFormatException e)
    {
        System.out.println("Invalid input for the
        second number. Please enter a valid input");
        return;
    }
    try {
        int result = num1 / num2 ;
        // May throw Arithmetic Exception (division by 0).
        System.out.println("Result: " + result);
    }
    catch (ArithmeticException e)
    {
        System.out.println("Error! Division by zero
        is not allowed");
    }
    catch (IOException e)
    {
        System.out.println("An input/output error
        occurred" + e.getMessage());
    }
}
}
}

```

Output:

Enter the first number : 4

Enter the second number: A

Invalid input for the second number. Please enter a valid integer.

Enter the second number: 0

Error! Division by zero is not allowed.